Module
Presenter's Manual

for

Working with MySQL

Effective from: June, 2024
Ver. 1.0

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	June 2024	New	-

Table of Contents

1. Introduction

At the end of this course, students will be able to:

- Understand basics of MySQL
- Understand various installation steps for MySQL
- Learn commands to create, view and alter a database and table
- Learn various commands to work with data
- Identify the different types of table joins in MySQL
- Use various functions in MySQL
- Create and use Stored Procedures
- Explain transactions and how to handle them
- Describe Performance Optimization
- Explain capability of MySQL for scaling and availability
- Explain replication in MySQL
- Perform data management using replication
- Understand concepts of partitioning in MySQL
- Understand concepts of Storage Systems and Management

2. Information on Session Allocation

Module	Online Hours
Working with MySQL	**24**

Throughout this Presenter's Manual, the module **Working with MySQL** will be referred to as **MySQL**.

3. Module Deliverables on OnlineVarsity

To aid the teaching process, following are the deliverables:

Resources available on Onlinevarsity:

Feature - Description/Functionality
E**book**-Student reading material in the form of PDF.
Download* *e**Book -** Student has the option to download the subject related e-book and read offline.
Practice 4 Me - Student can test and evaluate their understanding of module related topics.
Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
References - Student can access additional subject related material for reading.
Show Me How - Student can view a step-wise simulation/demonstration of the module related topics.
Glossary - Student can access a list of subject related specialized words with their definitions.
FAQ - Student can access frequently asked questions and their answers.

4. Week-wise Session Schedule

- **A session is of 2 hours duration.**

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 MySQL – TL1	Session 2 MySQL – TL2	Session 3 MySQL –TL3	Session 4 MySQL – TL4
2	Session 5 MySQL –TL5	Session 6 MySQL – TL6	Session 7 MySQL – TL7	Session 8 MySQL –TL8
3	Session 9 MySQL – TL9	Session 10 MySQL – TL10	Session 11 MySQL –TL11	Session 12 MySQL – TL12

MySQL: Working with MySQL
TL: Online Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	**MySQL – TL1**	All the topics as listed below from Session 1 of *Practical MySQL* book should be covered in this session. **Session 1 - Introduction to MySQL** ➢ Define database ➢ List and explain different elements of database management system ➢ Describe the principles of MySQL ➢ List different types of database models ➢ Define RDBMS and its terminology ➢ Establish MySQL Connection ➢ Outline the concepts of Database Design in DBMS ➢ Explain how to connect to Database Server Using MySQL Workbench ➢ Explain features, limitations, and deployment of MySQL ➢ Explain different elements of Normalization in DBMS	**Practical MySQL** SG-Session 1 XP-Session 1 TG-Session 1
2	**MySQL – TL2**	All the topics as listed below from Session 2 of *Practical MySQL* book should be covered in this session. **Session 2 – Essentials of MySQL** ➢ List MySQL Data Types ➢ List and explain different commands in MySQL ➢ Explain the basics of creating a database ➢ Identify ways to create and drop a database in MySQL ➢ Identify ways to select a database in MySQL ➢ Outline how to create, alter and drop a table	**Practical MySQL** SG-Session 2 XP-Session 2 TG-Session 2

<table>
<tr><th>Session No.</th><th>Session Title</th><th>Session Details</th><th>Deliverables' Mapping</th></tr>
<tr><td>3</td><td>MySQL – TL3</td><td>All the topics as listed below from Session 3 and Session 4 of Practical MySQL book should be covered in this session.

<u>Session 3 – Working with Data in MySQL</u>

➢ Explain the SELECT command in detail
➢ Identify the use of INSERT, UPDATE, and DELETE in the manipulation of data
➢ Outline the use of LIKE, IN, BETWEEN, and DISTINCT clauses
➢ Explain MERGE command and its usage

<u>Session 4 – Joins</u>

➢ Define JOINS
➢ Explain the use of JOINS
➢ Identify different types of JOINS
➢ Explain the use of SET operators
➢ Outline the differences between JOINS and UNIONS</td><td>Practical MySQL

SG-Session 3 & 4
XP-Session 3 & 4
TG-Session 3 & 4</td></tr>
<tr><td>4</td><td>MySQL – TL4</td><td>The Workshops of Session 1 to 4 of Practical MySQL course should be covered in this session.</td><td>Practical MySQL
(OnlineVarsity)
Session 1-4</td></tr>
<tr><td>5</td><td>MySQL – TL5</td><td>All the topics as listed below from Session 5 of Practical MySQL book should be covered in this session.

<u>Session 5 – Subqueries</u>

➢ Explain subqueries and their usage
➢ Outline the clauses and keywords used in subqueries
➢ Describe practical uses of WHERE and FROM clauses in subqueries
➢ Define IN, NOT IN, EXISTS, and NOT EXISTS keywords
➢ List and explain different types of subqueries</td><td>Practical MySQL

SG-Session 5
XP-Session 5
TG-Session 5</td></tr>
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Session No.	Session Title	Session Details	Deliverables' Mapping
6	**MySQL – TL6**	All the topics as listed below from Session 6 and Session 7 of *Practical MySQL* book should be covered in this session. **Session 6 – Functions and Stored Procedures** ➢ Explain the use of functions in MySQL ➢ Explain different types of functions ➢ Describe the usage of stored routines ➢ Explain how functions are created ➢ Outline the uses of stored procedures and functions **Session 7 – MySQL Clauses and Indexes** ➢ Explain the use of functions in Describe MySQL HAVING and ORDER BY clause ➢ Describe MySQL GROUP BY clause ➢ Define and use ROLLUP Modifier ➢ Explain Indexes in MySQL	**Practical MySQL** SG-Session 6 & 7 XP-Session 6 & 7 TG-Session 6 & 7
7	**MySQL – TL7**	The Workshops of Session 5 to 7 of *Practical MySQL* course should be covered in this session.	**Practical MySQL** (OnlineVarsity) Session 5-7
8	**MySQL – TL8**	All the topics as listed below from Session 8 of *Practical MySQL* book should be covered in this session. **Session 8 – Transactions, Performance Management and Backup** ➢ Define transaction ➢ List different commands used in transactions ➢ Describe transaction using JDBC Driver ➢ Outline different techniques to improve and manage database performance ➢ Explain Replication	**Practical MySQL** SG-Session 8 XP-Session 8 TG-Session 8

Session No.	Session Title	Session Details	Deliverables' Mapping
9	**MySQL – TL9**	All the topics as listed below from Session 9 and Session 10 of *Practical MySQL* book should be covered in this session. **Session 9 – Replication and Scalability** ➢ Explain MySQL replication architecture ➢ Develop and build on replication ➢ Describe replication topologies ➢ List replication use cases ➢ Summarize types of replications ➢ Identify advantages and disadvantages of MySQL replication ➢ **Session 10 – Partitioning** ➢ Define partitioning and its different types ➢ Describe how partitioning works ➢ List the benefits of partitioning ➢ Explain partitioning maintenance, pruning and selection ➢ Outline the restrictions and limitations of partitioning	**Practical MySQL** SG-Session 9 & 10 XP-Session 9 & 10 TG-Session 9 & 10
10	**MySQL – TL10**	The Workshops from Session 8 to Session 10 of *Practical MySQL* course should be covered in this session.	**Practical MySQL** (OnlineVarsity) Session 8-10
11	**MYSQL – TL11**	All the topics as listed below from Session 11 and Session 12 of *Practical MySQL* book should be covered in this session. **Session 11 – Performance Optimization in MySQL** ➢ Explain optimizing queries with MySQL query optimization guidelines ➢ Outline the process of optimizing workload ➢ Define how to monitor fundamental resources ➢ Describe restructuring queries ➢ Describe optimizing subqueries	**Practical MySQL** SG-Session 11 & 12 XP-Session 11 & 12 TG-Session 11 & 12

Session No.	Session Title	Session Details	Deliverables' Mapping
		➢ Identify scalability problems ➢ Explain queue cache ➢ Describe and explain automating of configuration ➢ Explain specific types of queries **Session 12 – Concept of Storage System and Management** ➢ Outline the types of data storage ➢ Explain different types of storage ➢ Describe demand paging and thrashing ➢ Explain process management in DBMS ➢ Define paging and segmentation	
12	**MYSQL – TL12**	The Workshops from Session 8 to Session 12 of *Practical MySQL* course should be covered in this session.	**Practical MySQL** (OnlineVarsity) Session 9-12

6. Library References

➢ **Learning MySQL** by Sergey Kuzmichev and Vinicius M. Grippa
➢ **Murach's MySQL** by Joel Murach
➢ **MySQL(TM): The Complete Reference** by Vikram Vaswani